

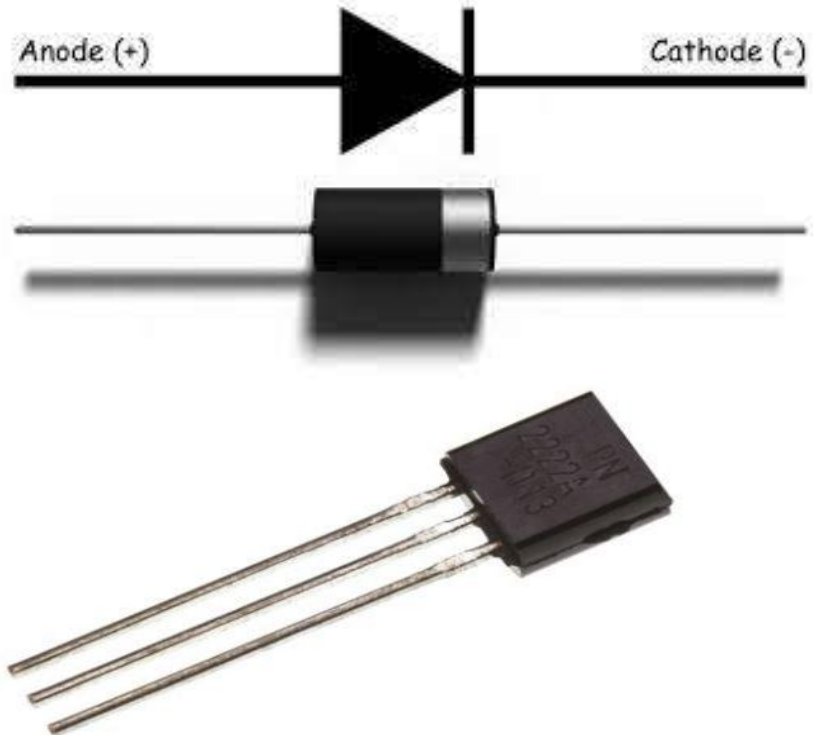
Logic Gates & Hardware

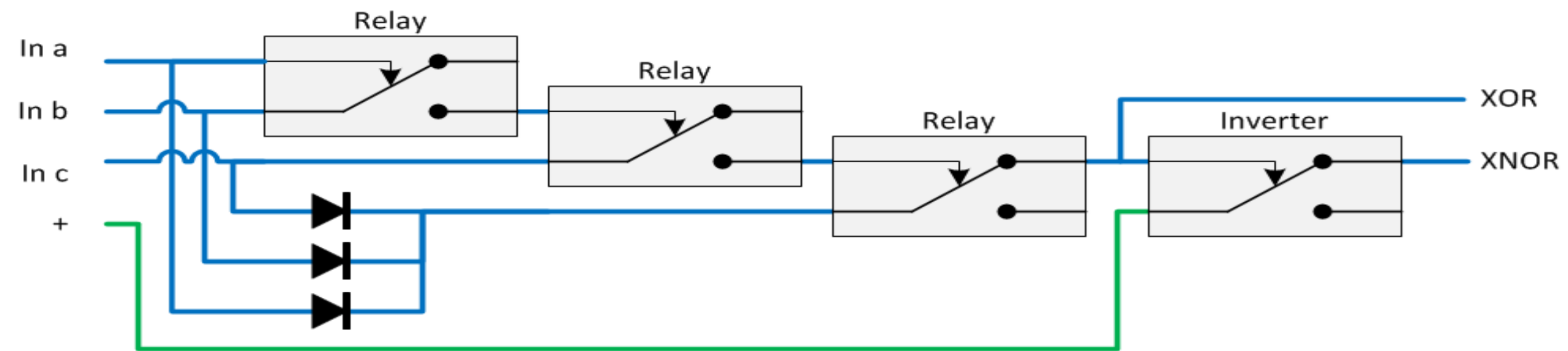
Evolution, NOT, OR & AND Gates

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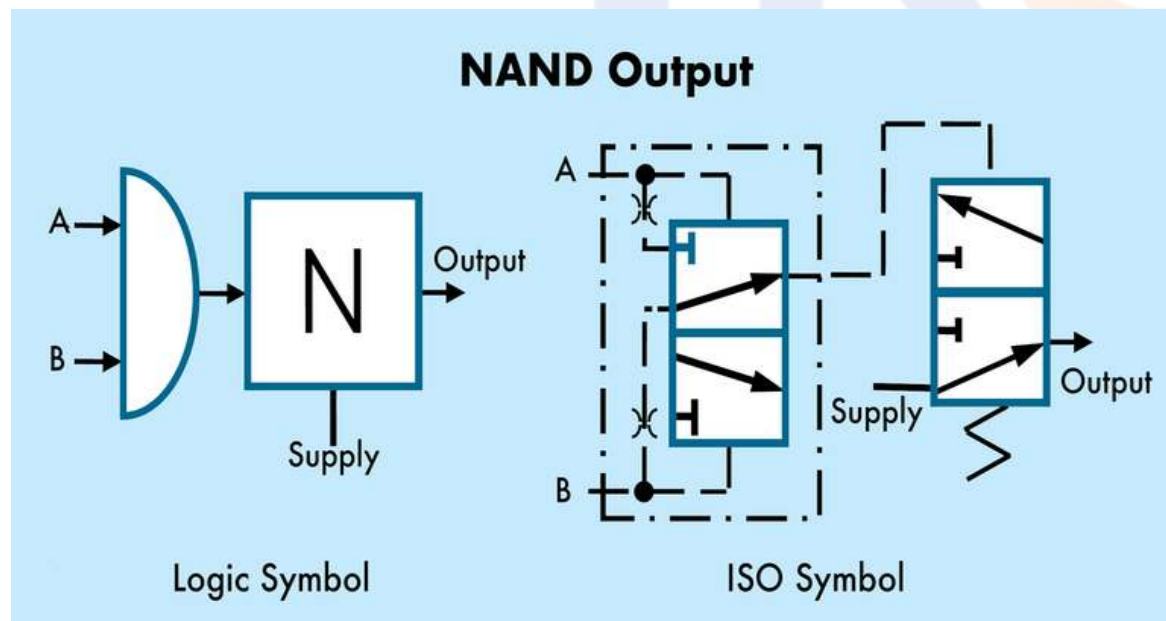
Logic gate

- In electronics, a logic gate is a physical device implementing a Boolean function. Logic gate performs a logical operation on one or more binary inputs signals and produces a single binary output signal. Logic gate is the basic building block from which many kinds of logic circuits can be constructed.
- Logic gates are primarily implemented using diodes or transistors acting as electronic switches, but can also be constructed using vacuum tubes, electromagnetic relays (relay logic), fluidic logic, pneumatic logic, optics, molecules, or even mechanical elements.





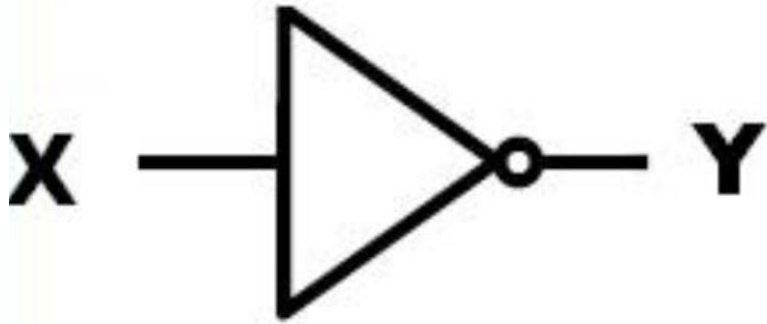
Electromagnetic Relays



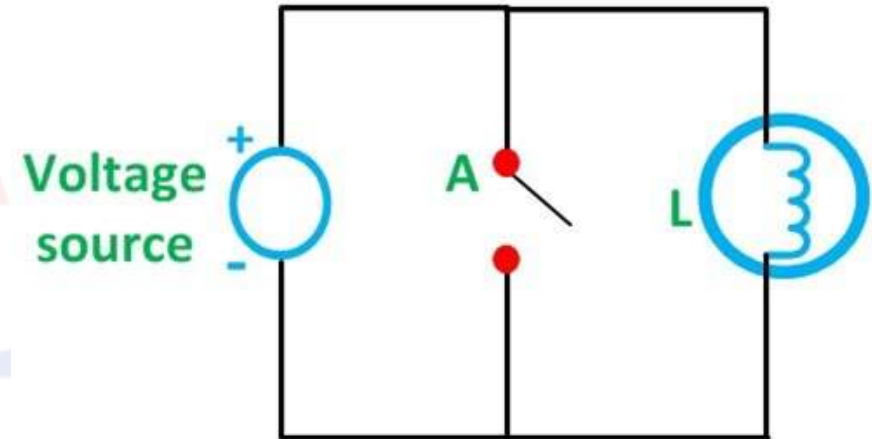
Pneumatic Logic

Not Gate(Inverter)

- It represents not logical operator is also known as inverter, it is a unary operator, which simply complement the input.

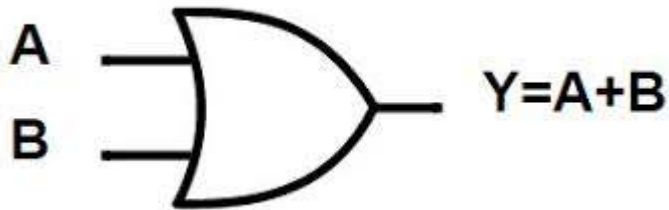


Truth Table	
Input	Output
X	$Y = X'$
0	
1	

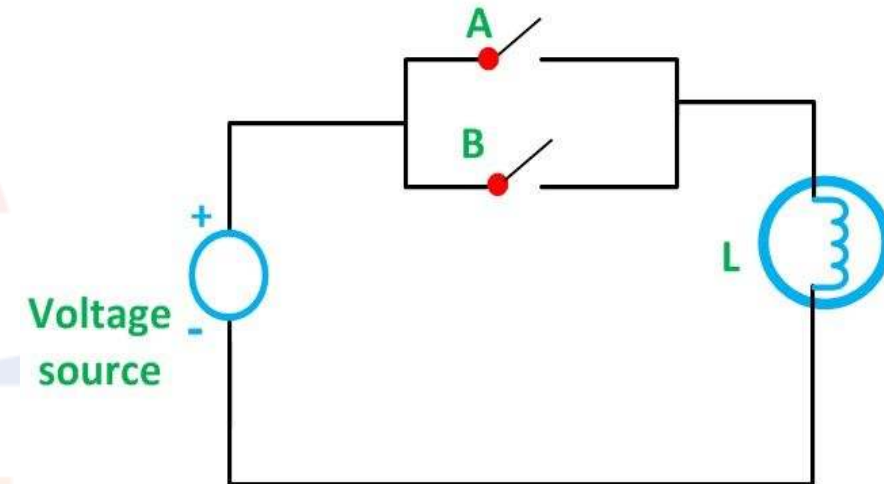


OR Gate

- It is a digital logic gate, that implements logical disjunction. The output will be high if at least one of the input lines is high.



Truth Table		
Input		Output
A	B	$Y = A + B$
0	0	
0	1	
1	0	
1	1	



- OR gate satisfy all the 3 rules idempotent, associative, and commutative.

1. Idempotent Law: $a + a = a$

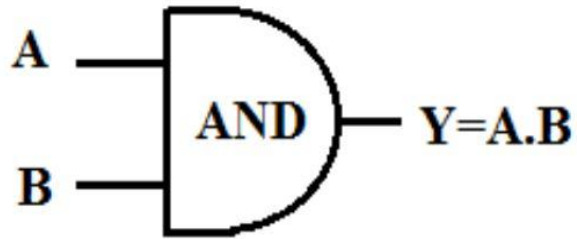
1. Associative law: $a + (b + c) = (a + b) + c$

1. Commutative law: $a + b = b + a$

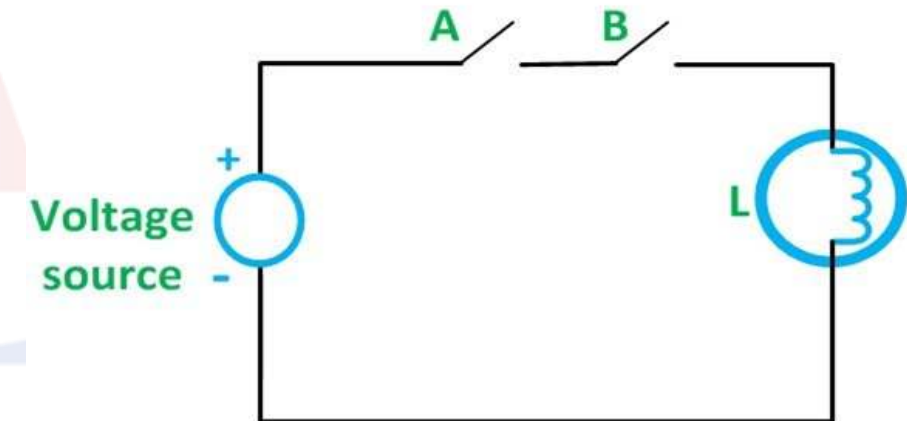
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And Gate

- It is a digital logic gate, that implements logical conjunction. Output will be high if and only if all input are high otherwise low.



Truth Table		
Input		Output
A	B	$Y = A \cdot B$
0	0	
0	1	
1	0	
1	1	



- And gate satisfy all the 3 rules idempotent, associative, and commutative.

1. Idempotent Law: $a \cdot a = a$

1. Associative law: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$

1. Commutative law: $a \cdot b = b \cdot a$

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Boolean Algebra Laws

Idempotent Law

$$a \cdot a = a$$

$$a + a = a$$

Associative law

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

$$a + (b + c) = (a + b) + c$$

Commutative law

$$a \cdot b = b \cdot a$$

$$a + b = b + a$$

Distributive law

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

$$a + (b \cdot c) = (a + b) \cdot (a + c)$$



De-Morgan law

$$(a + b)' = a' \cdot b'$$

$$(a \cdot b)' = a' + b'$$

Identity law

$$a + 0 = a$$

$$a \cdot 0 = 0$$

$$a + 1 = 1$$

$$a \cdot 1 = a$$

Complementation law

$$0' = 1$$

$$1' = 0$$

$$a \cdot a' = 0$$

$$a + a' = 1$$

Involution law

$$(a')' = a$$

